FIFTH SEMESTER (CUFYUGP) DEGREE EXAMINATION: MODEL PAPER 4

MATHEMATICS

ABSTRACT ALGEBRA I (MAT5CJ302)

Module 3: Cosets of a Subgroup, Lagrange's Theorem and Consequences

Maximum Time: 2 Hours		**Maximum Marks: 70**
	Section A	
	All Questions can be answered. Each Question carries 3 marks (Ceiling: 24 Marks)	
1	Let H be a subgroup of a group G. For $a, b \in G$, define $a \sim_R b$ if and only if $ab^{-1} \in H$. Prove that $\sim_R$ is an equivalence relation on G.	**3**
2	Write all the left cosets of the subgroup $3\mathbb{Z}$ in the additive group $\mathbb{Z}$, and explain why they form a partition of $\mathbb{Z}$.	**3**
3	Let $G = \mathbb{Z}_{12}$ under addition modulo 12, and let H be the subgroup $\langle 4 \rangle$. Find all distinct cosets of H in G and write down the index $(G : H)$.	**3**
4	Define the index of a subgroup H in a group G. Find: (a) the index of A_n in S_n for $n > 1$, and (b) the index of $5\mathbb{Z}$ in $(\mathbb{Z}, +)$.	**3**
5	Let H be a subgroup of a group G and let $a, b \in G$. Prove that if $aH = bH$, then $Ha^{-1} = Hb^{-1}$.	**3**
6	Let H be a subgroup of a group G. Prove that for any $a \in G$, the left coset aH has the same cardinality as H.	**3**
7	Prove that any group of prime order is cyclic and hence abelian.	**3**
8	Suppose G is a group of order 31. Can G have any subgroup other than $\{e\}$ and G? Up to isomorphism, how many groups of order 31 exist? Justify your answer.	**3**
9	Show that if a finite group G with identity e has order n, then $a^n = e$ for all $a \in G$. Hence calculate the remainder obtained when 45^{72} is divided by 73.	**3**
10	Does there exist a subgroup of a group of order 6 whose left cosets give a partition of the group into: (a) 6 cells, (b) 4 cells, (c) 12 cells? Give an example if possible, or explain why it is impossible.	**3**
	Section B	
	All Questions can be answered. Each Question carries 6 marks (Ceiling: 36 Marks)	
11	Let $G = \mathbb{Z}_{24}$ under addition modulo 24 and let $H = \langle [3] \rangle$. List the elements of H, find all the cosets of H in G, and state the index of H in $\mathbb{Z}_{24}$.	**6**
12	Let $\sigma = (1, 2, 5, 4)(2, 3)$ be a permutation in S_5. Express σ as a product of disjoint cycles, find the order of $\langle \sigma \rangle$, and find the index of the subgroup $\langle \sigma \rangle$ in S_5.	**6**
13	Find all the left cosets and all the right cosets of the subgroup $H = \langle \mu \rangle = \{\iota, \mu\}$ of the dihedral group $D_4 = \{\iota, \rho, \rho^2, \rho^3, \mu, \mu\rho, \mu\rho^2, \mu\rho^3\}$. Are the left cosets and right cosets of H the same? Justify.	**6**
14	Show that if H is a subgroup of index 2 in a group G, then every left coset of H is also a right coset of H. Hence prove that $g^{-1}hg \in H$ for all $g \in G$ and all $h \in H$.	**6**
15	Define the order of an element in a group G. Show that in a finite group G, the order of any element divides the order of G. Does every finite group contain an element of every order that divides the order of the group? Justify.	**6**
16	Show that every proper subgroup of a group G, with $o(G) = pq$ where p and q are prime numbers, is cyclic.	**6**
17	Suppose H and K are subgroups of a group G such that $K \leq H \leq G$, and suppose $(H : K)$ and $(G : H)$ are both finite. Prove that $(G : K)$ is finite and $(G : K) = (G : H)(H : K)$.	**6**
18	(a) Show that a group with at least two elements but with no proper nontrivial subgroups must be finite and of prime order. (b) Using the theorem of Lagrange, show that if n is odd, then an abelian group of order $2n$ contains at most one element of order 2.	**6**
	Section C	
	Answer any ONE. Each Question carries 10 marks (1x10=10 Marks)	
19	(a) Let H be a subgroup of a group G. Define the relation $\sim_L$ on G by $a \sim_L b$ if and only if $a^{-1}b \in H$. Prove that $\sim_L$ is an equivalence relation on G and show that the cell in this partition containing $a \in G$ is the left coset $aH = \{ah \mid h \in H\}$. (b) Find the partition of the group $\mathbb{Z}_6$,	**10**

	under addition modulo 6, into cosets of the subgroup $H = \{0, 3\}$. Construct the group table for $\mathbb{Z}_6$ with elements arranged in the order of these cosets and show that the coset blocks form a group isomorphic to $\mathbb{Z}_3$.	
20	(a) Let H be a subgroup of a finite group G. State and prove the Theorem of Lagrange showing that the order of H is a divisor of the order of G. (b) Use Lagrange's theorem to prove Euler's theorem: if $a, n \in \mathbb{Z}$ with $n > 0$ and $\gcd(a, n) = 1$, then $a^{\varphi(n)} \equiv 1 \pmod{n}$. Hence deduce Fermat's little theorem.	**10**